Week 2: Agentic RAG

- **What is Agentic RAG?**
  - Agentic RAG is an evolution of the traditional RAG framework that adds agents to the mix.
  - These agents add additional intelligence, including the ability to adapt, to plan, and to make decisions.
  - Agentic RAG allows LLMs to pull from multiple sources of data, allowing for more complex, multi-step workflows.
- **Traditional RAG works as follows:**
  - It involves connecting an LLM to some type of external knowledge base (like a vector database).
  - When a user asks a question:
    - A retriever retrieves the most relevant data.
    - The data is combined with the incoming user's query.
    - The LLM forms the response based on both.
- **What does Agentic AI mean?**

  Agentic AI has three main characteristics:

  - Memory:
    - Memory means the ability to recall previous behavior and perform future behavior based on that memory.
    - A memory can be created through semantic caching that allows it to cite previous queries and results.
  - Planning and Decision-making:
    - This is the ability to break down a task, make decisions about what to do, and adapt their behavior as the task goes along.
  - Tool use:
    - This is the ability to call APIs or tools to accomplish specific tasks in the context of a workflow.
    - It is even able to make the decision autonomously which tool to use.
- **Types of Agents in Agentic RAG**
  - **Routing Agents:**
    - Decide which tools and data sources to use for a query.
    - Guide queries to the appropriate part of the RAG system.
  - **Query Planning Agents:**
    - Break complex queries into smaller steps or sub-queries.
    - Distribute these steps to other agents and combine responses.
    - Act like project managers of the RAG pipeline.
  - **ReAct Agents (Reasoning + Action):**
    - Think step-by-step and act based on each result.
    - Adjust workflows on the fly using reasoning + tool use.

    - **Plan-and-Execute Agents:**
        - Build a plan and execute it autonomously.
        - No need to return to a controller after each step.
        - More efficient and often more accurate.
- **Advantages of Agentic RAG**
    - Dynamic tool usage (e.g., querying APIs, parsing emails, searching the web).
    - Improved handling of complex tasks and multi-step reasoning.
    - Greater accuracy through orchestration and self-adjustment.
    - Higher efficiency via autonomous execution.
- **Real-world Applications:**
    - AI agents handling complex customer queries across multiple systems.
    - Multi-step document analysis and summarization.
    - Dynamic business intelligence tasks using live APIs and databases.